

Ineffability as a window into hidden grammar in Laal

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Abstract

In this paper, I describe a case of ineffability—the impossibility to produce a morphologically complex form due to specific phonological conditions not being met—in Laal (isolate, Chad). Although the phonological conditioning of ineffability is straightforward and typologically unsurprising, the pattern is nonetheless particularly interesting because of the speakers’ response to the problem of ineffability. Indeed, when asked to derive such forms, speakers, far from simply failing to produce them, make every effort to apply “illicit” phonological operations to produce a form. I argue that the clear choices that speakers make among the many possible such operations they could apply reveals aspects of the phonological grammar of Laal that would otherwise go unnoticed or be misanalyzed.

Keywords: ineffability, morphology-phonology interface, marginal grammar, Laal

1 Introduction

Interactions between phonology and morphology have long been of interest to anyone studying the architecture of grammar.¹ Sharon Inkelas’ (2014) book on *The Interplay of Morphology and Phonology* offers one of the most exhaustive overviews of such interactions, their significance, and the various analyses they have given rise to. Among the many cases surveyed is one that is quite spectacular: *INEFFABILITY*, or the inability to apply “a word-formation construction, such as affixation, reduplication, or truncation” because specific phonological conditions on input or output fail to be met, leading to paradigm gaps (Inkelas 2014: 289). A well-known example is the English comparative suffix *-er*, which is compatible only with bases that are monosyllabic, e.g., *green-er*, *light-er*, or certain disyllabic ones with a stressless second syllable, e.g., *common-er*, *subtl(e)-er*. With longer bases, the comparative form cannot be derived (with very few exceptions), e.g., **brilliant-er*, **difficult-er*. In the case of English, a solution exists to

¹I owe Sharon Inkelas a huge debt of gratitude for her invaluable support throughout graduate school as my Ph.D co-advisor. It is thus an honor for me to be able to show some of this gratitude by contributing to this celebration of Sharon’s career and of the major impact she has had on the fields of phonology and morphology. Sharon’s introductory phonology course is what got me interested in phonology and theoretical modeling 15 years ago. Her research seminar on the interplay between phonology and morphology is where I discovered the very existence of ineffability. It is therefore only natural that I should contribute to this volume with a paper in which I propose to model an intriguing case of ineffability. May this serve as a demonstration of the long-lasting source of inspiration that Sharon has been and continues to be.

make up for this paradigm gap: longer adjectives lend themselves to an analytic comparative construction with the same meaning, e.g., *more brilliant*, *more difficult* (Inkelas 2014: 290). Other languages lack a default solution to remedy ineffability, and some forms are just not effable at all. This is seen, for example, in Turkish, where monosyllabic CV roots may not combine with suffixes consisting of one consonant only, because the output form fails to meet the requirement that suffixed words should be minimally disyllabic. For example, one may derive the passive form *sölje-n* ‘speak-PASS’, but not **de-n* ‘say-PASS’ (Inkelas 2014: 293).

In this paper, I describe a new case of phonologically-conditioned ineffability found in Laal, a language isolate of Southern Chad. What is particularly interesting in the Laal case, beyond the ineffability itself, is the way speakers behave when asked to derive the ineffable forms. Instead of simply failing to produce the relevant forms, they are first taken aback, hesitate, and then use various exceptional strategies (i.e., strategies seen nowhere else in the phonology of the language) in order to solve the ineffability issue and produce the supposedly underivable forms—although they always seem to be improvising and never produce the same form. The main point of the paper is not to propose a definitive account of ineffability in Laal, but to (i) clearly describe the phonological conditions leading to ineffability, and (ii) show that the *in-extremis* strategies speakers employ to solve the ineffability issue reveal aspects of the phonological grammar of Laal that would otherwise go unnoticed.² The analysis is couched in Optimality Theory (OT; Prince & Smolensky 1993, 2004), not because this formalism is crucial to the understanding of the phenomenon, but because it allows us to clearly identify and explore the hidden phonological properties revealed by ineffability.

After a few preliminary remarks on Laal phonology and morphology in §2, I first lay out the general phonotactic and prosodic well-formedness constraints at work in Laal in §3, before turning to the description of the ineffable cases in §4. Section 5 then gives a first OT analysis of the general constraints described in §3. Finally, §6 ventures into slightly unorthodox modeling to show how the speakers’ response to ineffability questions the analysis presented in §5 and reveals hidden peripheral aspects of the phonological grammar of Laal.

2 Preliminary remarks on Laal

2.1 Language and data

Laal is a language isolate spoken by ca. 800 speakers, mainly in the two villages of Gori and Damtar in southern Chad, as well as in towns such as Sarh, Bousso, and the capital N’Djamena. The data in this paper come from about 20 months of fieldwork on the language between 2010 and 2020, within a language documentation project funded by the DOBES (Documentation of Endangered Languages) project of the Volkswagen Foundation.³ The collected data, both elicitation sessions and spontaneous texts, is openly accessible in the Laal collection of the DOBES archive (Lionnet et al. 2020).

²For different ways in which ineffability can be accounted for, in particular in Optimality Theory, see Carstairs-McCarthy (1998); Orgun & Sprouse (1999); Baković & Keer (2001); Fanselow & Féry (2002); Beaver & Lee (2004); Raffelsiefen (2004); Smith (2013), among others; see also references in Inkelas (2014: 289–300).

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2.2 Preliminary phonological and morphological information

The vowel inventory is given in Table 1. Three levels of analysis are provided in the table: the underlying level which lists all contrastive vowels; the surface level, which corresponds to the inventory after application of phonological rules such as vowel harmony (see below); and the phonetic level, i.e., the actual realization of these vowels (only the typical realizations of each vowel is given). Throughout the paper, underlying representations will be given between slashes /.../, and surface representations will be in italics.

Table 1: Laal vowels

	Stem-initial (+length contrast)				Non-initial		
Underlying	/i/		/i/	/u/	/i/	(i)	/u/
	/e/		/ə/	/o/	/e/	(ə)	/o/
	/ɛ/		/a/	/ɔ/		/a/	(ɔ)
Surface	i	y	ɨ	u	i	ɨ	u
	e	ø	ə	o	e	ə	o
	ɛ	œ	a	ɔ		a	
Phonetic	[i]	[y]	[ɨ]	[u]	[i]	[ɨ]	[u]
	[e]	[ø~ʊo]	[ə]	[o]	[e]	[ə]	[o]
	[ɛ~ɨa]	[œ~ʊa]	[a]	[ɔ~ʊa]		[a]	

As can be seen from the stem-initial inventory on the left-hand side of Table 1, there are a total of 12 surface vowels in Laal, only nine of which have contrastive status. The front rounded vowels *y*, *ø*, and *œ* are indeed not contrastive, and always result from either rounding harmony or the realization of a floating round feature on a front vowel, e.g., /mín-ò/ → *mýnú* ‘ask-her’, /mêr-[-low,+round]/ → *môr* ‘star-PL’.

All vowels are attested in stem-initial position. Length is contrastive for all vowel qualities, in stem-initial position only. Vowel length is indicated in transcription by doubling the vowel, e.g., /ɛ:/ = /ɛɛ/. As seen on the right-hand side of Table 1, outside of the initial position, the underlying vowel inventory is reduced mostly to the five peripheral vowel inventory /i e a o u/, with /i/, /ə/ and /ɔ/ having only marginal contrastive status. The central vowels /ɨ/ and /ə/ are indeed contrastive in only a handful of morphemes (native morphemes for /ə/, only loanwords for /ɨ/). Surface *ɨ* and *ə* are mostly contextual allophones of other vowels (*ɨ* < /i/ or /ə/, *ə* < /e/ or /a/, respectively). As for non-initial /ɔ/, it is only present in the two object suffixes /-ón~-ó/ ‘you (SG)’ and /-òn~-òr/ ‘them (NT)’. This vowel is systematically changed to *a* on the surface, due to a ban on surface low peripheral vowels *ɛ* and *ɔ* outside of the stem-initial position. The underlying rounding of the vowel in these two suffixes is given away by the fact that they trigger anticipatory rounding harmony, e.g., /dāg-ón/ → *dǎgán* ‘drag-you.SG’, /dāg-òn/ → *dǎgàn* ‘drag-them.N’. Finally, The absence of the front rounded vowels *y*, *ø*, and *œ* outside of the stem-initial position is a direct consequence of the fact that they result from the application of regressive rounding harmony or the realization of a floating [+round] suffix, both of which affect only the the stem-initial syllable (see (5) below).

The low peripheral vowels ε and ɔ , as well as the non-high round vowels ø and œ have a very strong tendency to be realized as diphthongs, i.e., $[\varepsilon \sim \text{ia}]$, $[\text{ɔ} \sim \text{ua}]$, $[\text{ø} \sim \text{ɥo}]$, and $[\text{œ} \sim \text{ɥa}]$ respectively. Despite this diphthongized realization, they all behave like regular monomoraic vowels (bimoraic when long, in which case the phonetic diphthongization is by far the most frequent).

The $[\pm \text{low}]$ harmony pattern discussed in (4) shows that ε , œ , a , and ɔ are all phonologically low vowels. Laal thus has a rectangular vowel system, organized along a three-way height contrast. On the “horizontal axis”, the system opposes front and non-front vowels. This is shown by the rounding harmony discussed in (5), whereby the front vowels $/i \text{ e } \varepsilon/$ are rounded into front rounded y , ø , and œ respectively, and the central vowels $/i \text{ ə } \text{a}/$ into the back rounded vowels u , o , and ɔ . Both front and non-front vowels are thus characterized by a rounding distinction (which is contrastive only in the latter case). The vowel system can be described using the four binary features $[\text{high}]$, $[\text{low}]$, $[\text{front}]$, and $[\text{round}]$, as shown in Figure 1.

	[+front]		[−front]		
[+high]	i	y	ɨ	u	
[−high]	e	ø	ə	o	[−low]
	ε	œ	a	ɔ	[+low]
	[−round]		[+round]		

Figure 1: Vowel features

The consonant inventory is presented in Table 2. The underlying and surface inventories are almost identical, and thus not distinguished in the table. The glottal stop ʔ is a default epenthetic consonant inserted at the beginning of underlyingly vowel-initial stems in order to provide them with the required onset consonant, e.g., $/\text{àám}/ \rightarrow \text{ʔàám}$ ‘chief’. Indeed, no surface stem ever starts with a vowel.

Table 2: Laal consonants

Initial					Non-initial			
p	t	c	k	(ʔ)	p~b	t~d	c~ɟ	k~g
b	d	ɟ	g					
mb	nd	ɲɟ	ŋg					
ɓ	ɗ	ʔɟ						
	s		h		s			
m	n	ɲ			m	n	ɲ	ŋ
	r					r		
	l					l		
w	j				w	j		

As can be seen in Table 2, the non-initial consonant inventory is drastically reduced, just like vowels. Mostly, the rich plosive inventory found stem-initially is reduced elsewhere to one series of plosives underspecified for laryngeal features, realized as voiced when stem-internal,

mostly voiceless (and unreleased) when stem-final. The stem-initial C_1V_1 sequence can thus be defined as a position of prominence in Laal.

What I call the STEM is defined as a lexical root and any following suffixes. There are no prefixes in Laal. As a consequence, the stem is most of the time coextensive with the morphological WORD. This is not always the case however, e.g., with frozen compounds or reduplicative forms, which are multi-stem words, e.g., frozen compound $\{gàà\}\{gùn-ál\}$ ‘*Uraeginthus bengalus*, bird sp.’, or frozen reduplication $\{ɣì\}\{ɣèl\}$ ‘chameleon’ (where stems are enclosed in braces {...}). Stems in Laal are maximally disyllabic, as we will see in §3.3.

Laal is a tonal language, with a three-way high (H, e.g., $\acute{a}$) vs. mid (M, e.g., $\bar{a}$) vs. low (L, e.g., $\grave{a}$) contrast. The Tone Bearing Unit is the mora. Seven stem-level patterns are regularly attested, illustrated in (1).

- | | | | |
|-----|-----|--------------|------------------|
| (1) | H | <i>kúmá</i> | ‘type of basket’ |
| | M | <i>kūmā</i> | ‘hide’ |
| | L | <i>kùmà</i> | ‘medication’ |
| | HL | <i>gúbà</i> | ‘scalp ringworm’ |
| | LH | <i>gùmál</i> | ‘melon sp.’ |
| | HLH | <i>mérál</i> | ‘star’ |
| | LHL | <i>bùúrà</i> | ‘pigeon sp.’ |

Complex patterns involving a M tone (HM, MH, and LM—ML is unattested) are only found in a few function words and recent loans, and otherwise actively avoided. When morpheme concatenation would create such patterns, the M tone is systematically changed to L, as illustrated in (2) (cf. Lionnet 2022).

- | | | | |
|-----|----------|--------------|------------|
| (2) | /dāg-án/ | <i>dàgán</i> | ‘drag-him’ |
| | /dāg-àn/ | <i>dàgàn</i> | ‘drag-it’ |

Laal is characterized by a rich vowel harmony system, with three harmony processes. A non-initial mid vowel is changed to high by a preceding high vowel by virtue of perseverative [+high] harmony, as illustrated in (3). Anticipatory [$\pm$ low] harmony causes a non-high initial vowel to harmonize its [low] feature value with that of the following vowel, as shown in (4). Finally, anticipatory rounding harmony rounds the initial vowel of a stem whenever the following vowel is round, as seen in (5).

- | | | | | |
|-----|--|---|---------------|--|
| (3) | [+high] harmony, e.g., with the possessive suffix /-ər/ ‘my’ | | | |
| | /mīw-ər/ | → | <i>mīwīr</i> | ‘my liver’ (/ -ər/ ‘my’ → -īr) ⁴ |
| | /mīlā-ər/ | → | <i>mīlīl</i> | ‘my eye’ |
| | /mbūl-ər/ | → | <i>mbūlīl</i> | ‘my navel’ |

- (4) Low harmony

⁴The final /r/ of /-Vr/ suffixes is realized [l] when following a stem ending with /l/.

- a. with high-vowel suffix, e.g., *medio-passive* /-íɲ/: stem low → mid
- | | | | | |
|---------|-----------|---|--------|------------------------|
| /ɛ/ → e | /ʔéér-íɲ/ | → | ʔééríɲ | ‘be chosen’ |
| /a/ → ə | /màŋà-íɲ/ | → | màŋíɲ | ‘be gathered’ |
| /ɔ/ → o | /ʔɔŋ-íɲ/ | → | ʔòŋíɲ | ‘buy from one another’ |
- b. with mid-vowel suffix, e.g., /-ər/ ‘my’: stem low → mid
- | | | | | |
|---------|-----------|---|--------|-------------|
| /ɛ/ → e | /péér-ər/ | → | péérər | ‘my shin’ |
| /a/ → ə | /māl-ər/ | → | mələl | ‘my tongue’ |
| /ɔ/ → o | /bòg-ər/ | → | bògər | ‘my chin’ |
- c. e.g., with low-vowel suffix /-àr ~ -àn/ ‘it’ (OBJ): stem mid → low
- | | | | | |
|---------|------------|---|---------|-----------|
| /e/ → ε | /léérí-àn/ | → | léér-àn | ‘roll it’ |
| /ə/ → a | /cār-àr/ | → | cār-àr | ‘want it’ |
| /o/ → ɔ | /sór-àr/ | → | sór-àr | ‘find it’ |

(5) Rounding Harmony:

e.g., suffix /-ò ~ -òn/ ‘her’ (→ -ù ~ -ùn by virtue of [+high] harmony)

/i/ → y	/pīg-òn/	pīgòn	‘tie her’
/e/ → ɔ	/léér-òn/	lóóròn	‘wrap her’
/i/ → u	/pír-ò/	púrù	‘catch her’
/ə/ → o	/lār-ò/	lòrò	‘wait for her’

In some cases, it is impossible to determine the underlying value of the [low] feature of the initial vowel. This happens when a bound noun root requires a suffix in both the singular and the plural, and one of the two suffixes has a [+low] vowel (triggering [+low] harmony on the root vowel) and the other one a [-low] vowel (triggering [-low] harmony on the root vowel). This is the case, for example, with *miàn-án/mèn-ú* ‘fly (SG/PL)’, *jàm-ál/jəm-í* ‘tooth (SG/PL)’, or *ŋgòr-àl/ŋgòr-ó* ‘cry (SG/PL)’. In such cases, the vowel of the root is transcribed with a capital letter in underlying representation: front /E/ = e ~ ia as in /mĒn-án~ú/, central /A/ = ə ~ a as in /jÀm-ál~í/, and back /O/ = o ~ ua as in /ŋĠÒr-àl~ó/.

These harmony rules (as well as other phonotactic constraints, e.g., *VVCC described in §3.2) reveal the existence of two distinct morphophonological strata in Laal. Stratum 1 consists in morphophonological processes taking place within roots and between roots and two types of suffixes: number-marking suffixes on nouns and verbs and the deverbal and denominal suffix /-VI/. Stratum 2 morphophonology is found with all other suffixes, including object suffixes, which are the most relevant for the case of ineffability described in this paper. The difference between the two strata is clearly shown with the behavior of rounding harmony, which applies unconditionally in stratum 2, but under very strict conditions in stratum 1: the target vowel must be in the vicinity of a labial consonant, and the trigger and target vowels must be of equal height and frontness specifications (cf. Lionnet 2017).

Since ineffability emerges from the concatenation of verb roots and object suffixes, it is important at this point to briefly introduce these suffixes. As seen in Table 3, there are two series of object suffixes: the N-series, with suffixes involving the consonant /n/, and the R-series, with suffixes (mostly) involving the consonant /r/. The latter are less frequent and less productive, and are probably older. ⁵

⁵Note that two other allomorphs are ignored here, for the sake of simplicity: (i) the special forms used

Table 3: Object suffixes

		Simple Object		Object+Associative	
		N-series	R-series	N-series	R-series
SG	1	-ón	-ór	-nó~-ní	-ró~-rí
	2	-ón	-ó	-nó	-ró
	3M	-án	-ár	-ná	-rá
	3F	-òn	-ò	-nó	-ró
	3N	-àn	-àr	-ná	-rá
PL	1EX	-nùrú~-nũ	-nùrú~-rũ	-nú	-rú
	1IN	-nǎŋ	-rǎŋ	-nárŋ	-ráŋ
	2	-nũŋ	-rũŋ	-núŋ	-rúŋ
	3MF	-nìrí~-nĩ	-nìrí~-rĩ	-ní	-rí
	3N	-òn	-òr	-nó	-ró

3 Phonotactic and prosodic well-formedness in Laal

I now turn to the description of three well-formedness constraints that play a crucial role in the case of ineffability that is the focus of this paper: a ban against triconsonantal clusters (*CCC) which is the result of strict constraints on syllable structure (§3.1), a ban against *VVCC sequences within stratum 1 morpho-phonology (§3.2), and a ban against stems longer than two syllables (§3.3).

3.1 Syllable structure well-formedness and *CCC

Attested syllable shapes are CV, CVC, CVV, and CVVC (where VV stands for a long vowel). V-initial stems systematically undergo initial glottal stop epenthesis to conform to the necessity of the presence of an onset, e.g., /ààm/ → ?ààm ‘chief’. Vowel hiatus is not tolerated, and diphthongs are unattested (the diphthongization of monomoraic ε , γ , $\emptyset$, and $\varnothing$ mentioned above is purely phonetic).

When a V-initial suffix is added to a CV(V) root, vowel hiatus is avoided in two partly unpredictable ways. With a random subset of *Ca(a)* verbs, hiatus is avoided by *r*-epenthesis, as seen in (6). In all other cases, vowel coalescence applies, as illustrated with combinations involving roots ending in *a(a)* in (7). Note that the result of coalescence is always a long vowel, even when the root vowel is already long (no extra-long vowel is created). In other words the length contrast is neutralized under coalescence, as seen by comparing the two forms in (7a-d).

- (6) a. /páá-àn/ *pááràn* ‘knock it down’
 e. /ʔjá-àr/ *ʔjáràr* ‘take it’

with monosyllabic /Ci/ and /Cu/ roots (e.g., /kú-rír/ ‘see-me’ and /bí-ní/ ‘bury me’), and (ii) the combined Object+Ventive forms, whose singular and third person neuter plural forms are segmentally identical to the combined Object+Associative forms in Table 3, and whose plural forms (except 3N.PL) are unattested and replaced by the simple forms.

- (7) a. a+ə /cá-ən/ céén ‘thrash-me’
 /ɲáá-ér/ ɲéér ‘leave me’
 b. a+o /cá-òn/ cóòn ‘thrash-her’
 /ɲáá-ò/ ɲóò ‘leave her’
 c. a+a /cá-án/ cáán ‘thrash-him’
 /ɲáá-ár/ ɲáár ‘leave him’
 d. a+ɔ /cá-ón/ cóón ‘thrash-you sg’
 /ɲáá-ó/ ɲóó ‘leave you sg’

A consequence of these constraints on syllable structure is that the maximum number of consonants per cluster is two: one coda followed by one onset. We will see in §4 that the impossibility to have tri-consonantal clusters plays an important role in the ineffability case described in this paper.

3.2 Stratum-1-internal *VVCC

VVCC sequences (i.e., a long vowel followed by a consonant cluster) are banned within stratum 1 (*VVCC). This ban is a strong static constraint with only two exceptions, both recent loanwords: *māārtā* ‘eight’ (< Barma) and *páàṅgú* ‘trousers open on the side’ (< Sango).

*VVCC is also an active constraint that knows no exception: whenever stratum-1 morphology would produce such a sequence, the long vowel is systematically shortened: *VVCC > VCC. This is illustrated below with examples where the combination of CVVC(V) noun roots with C-initial number-marking suffixes would otherwise create the unlawful VVCC sequence.⁶

- (8) a. /pĒĒr-ɲā/ pĒr-ɲā (*pĒĒr-ɲā) ‘snake-PL’ (cf. sg /pĒĒr-i/ pĒr-i)
 b. /māār-gā/ mār-gā (*māār-gā) ‘beer-PL’ (cf. sg /māār/ mār)
 c. /kōōg-mī/ kōg-mī (*kōōg-mī) ‘bone-PL’ (cf. sg /kōōg/ kōg)

VVCC sequences derived in stratum 2 are, however, allowed. These always involve /CVVC(V)/ roots followed by a /-CV(C)/ stratum-2 suffix, as briefly illustrated in (9) with the object suffix /-nũ/ ‘us (EX)’.

- (9) /dāār-nũ/ dōòr-nũ ‘make us (EX) enter’
 /jóólí-nũ/ jóól-nũ ‘turn around us (EX)’

3.3 Disyllabic maximum

As seen in Table 4, 98% of stems in Laal are either monosyllabic (43%) or disyllabic (55%). The remaining 2% consist of 49 trisyllabic stems and 1 tetrasyllabic, almost all recent loanwords, as illustrated in (10).⁷

⁶The vowel changes in the examples are caused either by regular anticipatory [±low] harmony, as in (8a) (cf. (4)), or by morpheme-specific metaphony: /-ga/ raises the low vowel of the root to [+high] in (8b), and /-mī/ lowers the root vowel to [+low] in (8c).

⁷Chadian Arabic words are taken from Jullien de Pommerol (1999), Barma words from Keegan & Idris (2016). There are only five trisyllabic stems that I was not been able to identify as loans: *mābúgná* ‘frog, toad (PL)’ (cf. sg *mābúg*, likely a frozen compound *mā+búg*, since it displays an illicit MH tone pattern), *gōšīrgi* ‘bird sp.’ (onomatopoeic?), *wòtíná* ‘type of whistle made of metal’, *dīngiri* ‘fish larva, baby fish’, and *jīmijà* ‘type of fishing net’.

Table 4: Stems by number of syllables

# syllable(s)	# stems	%
1 syllable	1159	43%
2 syllables	1499	55%
3 syllables	49	< 2%
4 syllables	1	< 1%
Total	2707	

- (10) a. *gùmáji* ‘clothes’ (< Arabic *gumâji*)
kàdàrbō ‘blue’ (< Barma *kàtàrpō*)
bìskilêd ‘bicycle’ (< French *bicyclette*)
b. *òrdìnàtêr* ‘computer’ (< French *ordinateur*)

The disyllabic maximum is both a static pattern and an actively enforced constraint. When the addition of a suffix to a disyllabic root would create a trisyllabic stem, the stem is shortened by reducing the second syllable of the root. This is achieved with minimal loss of root material by changing the vowel nucleus of the second syllable into a glide whenever possible, or by deleting the vowel when prevocalic gliding is impossible. Note that suffixal material is always entirely preserved, and only the second syllable of the root is affected.⁸ In the remainder of this section, I will describe the stem shortening processes seen with roots combining with V-initial suffixes. C-initial suffixes will be addressed in the next section, together with the cases of ineffability they sometimes generate. Illustrative examples only involve stratum-2 morphology, i.e., verb roots followed by verbal suffixes. Stratum-1 morphology shows exactly the same effects with respect to stem shortening, but is less regular in other respects, and better left aside for the sake of clarity.

prevocalic gliding is seen with CVCV_[+high] roots followed by a vowel-initial suffix, as shown in (11).⁹

- (11) /kōrī-àn/ *kòrj-àn* ‘decorate it’
/lóbí-àn/ *lábj-àn* ‘hide it from view’
/pèdí-àn/ *pàdj-àn* ‘fan it’
/bìrú-àn/ *bìrw-àn* ‘burn it’

There are only three exceptions to prevocalic gliding, listed in (12). In all three cases, V2 is deleted rather than changed into a glide.

⁸With only a few exceptions, cf. footnote 15.

⁹Remember that the only two high vowels attested outside of the stem-initial position in native words are /i/ and /u/. Note that the tone pattern of a base of suffixation in Laal is always reduced to its initial tone. This can be seen in the examples in this section. The deletion of non-initial tones is not due to V2 gliding or deletion, as seen by the following example: /sǎŋ-àn/ *sàŋ-àn* (**sǎŋ-àn*) ‘fight-it’. This example also shows that tone pattern simplification is not caused by any restriction on the stem-level tone pattern, since the LHL pattern that would obtain in the absence of tone deletion is a regularly attested stem pattern.

- (12) a. CVwi
 /jíwí-àr/ *jíw-àr* ‘forget it’
 /gíwí-àn/ *gíw-àn* ‘turn it upside down’
 b. CuCu
 /gūrū-àn/ *gūr-àn* ‘eat it (mush)’

All three are verbs that fall into two categories exhaustively represented in (12): the only two verbs ending in /wi/ (12a), and the only disyllabic verb with two high back rounded vowels (12b). Their segmental makeup offers a plausible explanation for the application of V2 deletion rather than prevocalic gliding. In the latter case, prevocalic gliding would yield the form **gùrwàn*, in which the glide would likely be interpreted as V1 leakage into V2. Indeed, [u] has a very strong tendency to leak through a following consonant, e.g., *jùgà* = [jùg^uà] ‘inside’. The form **gùrwàn* is thus very likely to be interpreted as **[gùr^uàn]*, and hypocorrected to *gùràn*.

Explaining the lack of prevocalic gliding in CVwi verbs is less straightforward. One can simply note that prevocalic gliding in this case would yield the cluster /wj/ (/jíwí-àn/ → **jújàn*), which is only attested once in the lexicon: *tāwj-ā*, PL of *tàwí* ‘fish sp.’ – although this one attestation originates precisely from the prevocalic gliding of a stem-final /i/. Furthermore, the two verbs in (12a) have a high non-back V1; prevocalic gliding would thus create a *iwj* or *úwj* sequence, both of which are unattested in the lexicon.

With all other vowel-final root shapes combining with a V-initial suffix, prevocalic gliding would violate at least one exceptionless phonotactic constraint, and is thus avoided in favor of full V2 deletion. This is the case with CVVCV_[+high] roots, where prevocalic gliding would create a stratum-1-internal VVCC sequence, as shown in 13.¹⁰

(13)	Gliding > *VVCC	V2 deletion	
/súúrí-àn/	<i>*súúrj-àn</i>	<i>súúr-àn</i>	‘change it’
/léérí-àn/	<i>*léérj-àn</i>	<i>léér-àn</i>	‘roll it’
/núúlí-àn/	<i>*núúlj-àn</i>	<i>núúl-àn</i>	‘swallow it’
/jóólí-àn/	<i>*jóólj-àn</i>	<i>jóól-àn</i>	‘turn around it’

With CVCCV_[+high], prevocalic gliding would create a triconsonantal cluster, strictly banned and actively avoided as we saw above. Once again, the disyllabicity constraint is satisfied by deleting the second root vowel, as illustrated in (14).¹¹

(14)	Gliding > *CCC	V2 deletion	
/dígrí-àn/	<i>*dígrj-àn</i>	<i>dígr-àn</i>	‘fool it’
/dǎgní-àn/	<i>*dǎgnj-àn</i>	<i>dǎgn-àn</i>	‘put (sth) against it’
/ngónrí-àn/	<i>*ngónrj-àn</i>	<i>ngónr-àn</i>	‘coil it’

With CVCVC roots, V2 gliding would likewise be problematic since it would yield a triconsonantal cluster. Once again, V2 deletion applies instead, as shown in (15).¹²

¹⁰All CVVCV_[+high] verbs are of shape CVVLi (L= liquid). The only exception is *méém-í* ‘be.bitter-PL’, which is a suffixed plural form, and is thus incompatible with any further suffixation.

¹¹There are no CVCCu verbs in the lexicon.

¹²There are no CVCiC verbs, except verbs of CVC-ijp shape, with the plural suffix /-ijp/ or the mediopassive suffix /-ijp/, which cannot be subject to further suffixation.

(15)		Gliding > *CCC	V2 deletion	
	/múnúŋ-àn/	*múnwŋ-àn	múnŋ-àn	‘give birth to it’
	/tūrūg-àn/	*tūrūg-àn	tūrūg-àn	‘sew it’
	/ŋgògùm-àn/	*ŋgògwm-àn	ŋgògm-àn	‘bend it’

When the second vowel of the root is non-high, prevocalic gliding is impossible—only peripheral high vowels can be changed into the corresponding glides *j* and *w*. In this case, V2 deletion applies systematically, irrespective of the shape of the root, as seen in (16).¹³

(16)	a.	CVCV-V....		
		/gùmà-ér/	gùm-îr	‘call me’
		/dēlē-uàn/	dæl-àn	‘send them back home’
	b.	CVVCV-V...		
		/núúlá-én/	núúl-în	‘swallow me’
		/sèègé-én/	sèèg-én	‘accuse me’
	c.	CVCCV-V...		
		/dìbgà-én/	dìbg-în	‘frighten me’
		/pilmé-àn/	pilm-àn	‘film it (<French <i>filmer</i>)’
	d.	CVCVC-V...		
		/lèlág-én/	lèlg-én	‘shake me laterally’
		/tègór-àn/	tàgr-àn	‘sift it’
		/cògór-àn/	cògr-àn	‘filter it’

The only regular exceptions to the disyllabic maximum are verb forms with a disyllabic object suffix: either 1PL.EX.OBJ /-nùrú/ or 3MF.PL.OBJ /-nirí/. Such trisyllabic forms as /sór-nùrú/ *sórñùrú* ‘find us (EX)’ and /sór-nirí/ *sórñirí* ‘find them’ are totally acceptable. It is interesting to note, however, that both suffixes have monosyllabic allomorphs: /-nũ/~/-rũ/ and /-nĩ/~/-rĩ/ respectively. Disyllabic *sórrũ* ‘find us (EX)’ and *sórrĩ* ‘find them’ comply with the disyllabic maximum constraint, and are as frequent as their trisyllabic variants.

Note, finally, that in the case of V-final roots combining with V-initial suffixes, another phonotactic constraint emerging from general constraints on syllable structure well-formedness would also be violated should prevocalic gliding or V2 deletion not apply: the ban on onsetless syllables (and on the resulting vowel hiatus), e.g., /pàdí-àn/ → *pə.dì.àn (instead of actual *pàdjàn* ‘fan it’, cf. (11)).

To summarize, it seems like the grammar strives to preserve every segment of the derived form while abiding by the disyllabic maximum constraint. The second syllable appears to be the weakest one, systematically targeted when shortening must apply. The form of syllable reduction that preserves as much of the underlying material as possible is V2 gliding, which involves only a feature value change from [+syllabic] to [–syllabic]. This only applies if V2 is a peripheral high vowel (/i/ or /u/), and if the resulting glide does not result in a triconsonantal cluster or a stratum-1-internal VVCC sequence. In all cases where prevocalic gliding is impossible, either because the resulting form would violate another phonotactic constraint (*CCC or *VVCC) or

¹³There are 167 disyllabic verbs (+ 54 plural forms, usually incompatible with any further suffixation, hence excluded from this count), the most frequent V2 is /a/ (82), followed by /i/ (49), /u/ (12), /ə/ (11), /o/ (7), and /e/ (6).

because V2 is not a high vowel, the vowel of the second syllable is entirely deleted. No other material is deleted—in particular, suffixal material is never affected by syllable reduction, including with disyllabic suffixes, which may exceptionally violate the maximal disyllabicity constraint, as seen above.¹⁴

4 Ineffable forms

4.1 C-initial suffixes cause ineffability

Let us now turn to combinations with C-initial suffixes, and the ineffability problem that they sometimes give rise to.

If the addition of a C-initial suffix does not create a triconsonantal cluster, then regular V2 deletion applies, as expected. This is seen only in CV(V)CV-C... cases, i.e., where the root is both V-final and is not preceded by a consonant cluster, as illustrated in (17) with the 1PL.INCL object suffix /-nǎŋ/. (Note that this suffix unexpectedly triggers [-low] harmony on the root vowel despite having a [+low] vowel, as seen in (17b) /pèdí-nǎŋ/ → pǝdnǎŋ, not *pàdnǎŋ.)

- (17) a. /nírǎ-nǎŋ/ *nír-nǎŋ* ‘leave us (INCL)’
 /gùmǎ-rǎŋ/ *gùm-rǎŋ* ‘call us (INCL)’
 b. /pèdí-nǎŋ/ *pǝd-nǎŋ* ‘fan us (INCL)’
 /birú-nǎŋ/ *bír-nǎŋ* ‘burn us (INCL)’

Prevocalic gliding of the high V2 in (17b) is avoided in favor of V2 deletion because it would create a triconsonantal cluster (e.g., /pèdí-nǎŋ/ → *pǝdjǎŋ).¹⁵

All other disyllabic roots contain two non-initial consonants: CVCCV and CVCVC. V2 deletion is not an option in these cases, because it would create a CCC cluster, as shown in (18).

- (18) /jùgár-nǎŋ/ **jùgr-nǎŋ* ‘shake us (INCL)’
 /dígri-nǎŋ/ **digr-nǎŋ* ‘fool us (INCL)’

It turns out that the grammar of Laal does not have a solution to this problem. Indeed, it seems that V2 deletion is already the last-resort solution to violations of the disyllabic maximum constraint when gliding is impossible. When even V2 deletion is impossible (because it would create a CCC sequence), the language is out of repair strategies. This was accidentally revealed to me during an elicitation session focusing on verb paradigms, as I was asking a speaker to produce these forms. I was able to verify later that the effect was the same with a handful of other speakers on different occasions. When asked to derive such forms, speakers are at first disconcerted, then hesitate between different solutions, all of them involving either violations of otherwise inviolable phonotactic constraints or illicit (i.e., not regularly attested) operations in order to comply with said constraints, before settling for one of these solutions—but not always the same one.

One invariable element in all the forms between which speakers hesitate is that whenever something is deleted to comply with stem well-formedness constraints, it is always material at

¹⁴Cf. footnote 15 for another minor exception.

¹⁵ In the case of CVCi-CV sequences, V2 deletion is in free variation with another strategy consisting in deleting the initial consonant of the suffix and gliding V2: /pèdí-nǎŋ/ = pǝdnǎŋ ~ pǝdjǎŋ ‘fan us (INCL)’.

the end of the base of affixation, or (more rarely) at the very beginning of the suffix, but never material at the beginning of the stem. The weakest element of the base is always V2, but deletion of up to the entire second syllable is also exceptionally attested in order to solve the ineffability problem. This follows from the fact that all of the attested alternations involve repairs at the root-affix boundary, and is in keeping with the prominent status of the stem discussed in §2.2.

I will illustrate this with data collected from Speaker AK1 during one single elicitation session on March 18, 2014 in Gori. Recordings of this elicitation session can be found in the Laal DoBeS archive, under the codes 140318-23-AK1 and 140318-24-AK1.¹⁶ I will content myself with the limited number of forms produced by AK1 during this elicitation session (exhaustively listed in Tables 5 and 6), because it is the only one that I recorded. I noticed the same behavior from other speakers (producing other similar verb forms) in other sessions that I unfortunately did not record. This allows me to hypothesize that this behavior generalizes to the entire community of speakers.

During this elicitation session, I asked AK1 to produce the three verbs /jùgár/ ‘to shake’, /págàr/ ‘to think (of)’, and /múnún/ ‘to give birth (to)’ in various forms involving a C-initial suffix: the object plural suffixes 1PL.EX.OBJ /-nùrú/~/~nũ/~/~rũ/, 1PL.IN.OBJ /-nǎŋ/, 2PL.OBJ /-nũŋ/, and 3MF.PL.OBJ /-nìrí/~/~nĩ/~/~rĩ/, as well as object+associative fused suffixes, all of shape /-nV/ (cf. Table 3). As an illustration, I give in (19) and (20) below a transcription of AK1’s answers when asked to produce the forms /jùgár-nìrí~nĩ~rĩ/ ‘shake-them’ and /múnún-nùrú~nũ~rũ/ ‘give.birth.to-us(EXCL)’, to give a sense of his reaction. An English translation of the sentence elicited in French is given in (a), the underlying form of the expected response in (b), and the actual response in (c) (hesitations are marked with “...”).

- (19) a. ‘We (EXCL) shake them.’
 b. /ùrú jùgár-nìrí~nĩ~rĩ/
 We.EX.SBJ shake-3MF.PL.OBJ
 c. *ùrú jùgr-rĩ... ùrú jùgr-nĩ... ùrú jùgr-n... ùrú jùg-nìrí... [long pause] ùrú jùgr-rĩ... ùrú jùgr-rĩ* (140318-24-AK1)

- (20) a. ‘My brother and I, we were born in Damtar.’¹⁷
 b. /já jà nèm ì múnún-nùrú~nũ~rũ buāl/
 I and my.brother they gave.birth.to-us in.Damtar
 c. *já jà nèm ì m... ì mún... ì múnún... ì múnú-ùn... ì múnú-nùrú buāl... ì múnú-nùrú buāl* (140318-23-AK1)

Based on the various forms produced by AK1 during this elicitation session, one can identify possible strategies for each CVCVCV form, involving various combinations of the four processes listed in (21), three of which are “illicit”, i.e., otherwise not seen anywhere else in the language. Note that here and in what follows, I use subscripted letters as indices to identify the three consonants in a triconsonantal cluster ($C_a C_b C_c$) in order to avoid any confusion with the

¹⁶A concordance table between these codes and the actual file names in the online archive can be found in Lionnet (2020).

¹⁷Note that the *n* in the cluster *ng* in the forms *múnú-ùn* and *múnú-nùrú* in (20) is a coronal nasal (the velar nasal is systematically transcribed *ŋ*, even in the prenasalized stop *ŋg*).

position of the consonants within the stem, which is indicated with a number (e.g., C_a is not C_1 , the stem-initial consonant, but always the second consonant in the stem, i.e., C_2).

- (21) a. V2 deletion (licit, expected)
 b. C_b deletion in $C_a C_b C_c$ clusters (illicit)
 c. C_c deletion in $C_a C_b C_c$ clusters (illicit)
 d. V2 epenthesis (illicit)

V2 deletion (21a) applies systematically in all forms, as expected. C-deletion strategies (21b) and (21c) apply in order to reduce the triconsonantal cluster created by V2 deletion, as shown in (22).

$$(22) \quad \begin{array}{ccccccc} & & \text{V2Del} > *CCC & & C_b \text{ deletion} & & C_c \text{ deletion} \\ /jùgár-nũ/ & \rightarrow & *jùgr-nũ & \rightarrow & jùg-nũ & \sim & jùgr-ũ \end{array}$$

Note that a form like $/pág_1àr_2-r_3ũ/ \rightarrow [pògrũ]$, where the root-final and the suffix-initial consonants are identical, could be analyzed in three ways: (i) $/póg_1r_3ũ/$ with C_b deletion, (ii) $/póg_1r_2-ũ/$ with C_c deletion, and (iii) $/póg_1r_{2,3}ũ/$, with fusion of C_b and C_c . For the sake of simplicity, I will exclude fusion from the list of strategies employed by speakers and consider that C_b deletion and C_c deletion are sufficient to account for the facts.

V2 epenthesis (21d) is another strategy employed to break the triconsonantal cluster created by the prior application of V2 deletion. However, this strategy reintroduces the initial problem solved by V2 deletion, by creating a trisyllabic form again. Interestingly, keeping the original V2 by simply not applying V2 deletion is never chosen as a strategy. That the vowel chosen as a repair in such cases is indeed epenthetic and not the original V2 is shown by the fact that the reintroduced V2 is not identical to the underlying V2, but always a copy of V1, as shown in (23a). In addition, the epentheticized vowel does not always occupy the same slot in the word as the originally deleted V2, as seen in (23b), where the original form is CVCVCV-CV, but the final form after V2 deletion and (re-)epenthesis is CVCCV-CV (both the original V2 and the epenthetic V2 are underlined; the change of η to g in $/múnún\eta-ní/ \rightarrow múnúgú-ní$ in (23b) is discussed below).

$$(23) \quad \begin{array}{ccccccc} & & \text{V2 del.} > *CCC & & \text{V2 ep.} & & \\ \text{a. } /jùgár-ná/ & \rightarrow & *jùgr-ná & \rightarrow & jùg\underline{ù}r-ná & \text{'shake him (ASS)'} \\ \text{b. } /múnún\eta-ní/ & \rightarrow & *mún\eta\eta-ní & \rightarrow & múnú\underline{g}ú-ní & \text{'give birth to me (ASS)'} \end{array}$$

Tables 5 and 6 present all the forms given by AK1 during the elicitation session. Forms that were not produced by AK1 are indicated with “—”. This does not mean that AK1 rejected or refused to use these forms, but simply that they were not among the forms that he spontaneously produced and hesitated between. I assume he could have chosen these other alternatives as well. Note that there is a wider variety of forms given for the VERB-object combinations in Table 5 than for the VERB-object:ASS combinations in Table 6. It is unclear what the reason behind this difference is. It is possible that by the time I started eliciting the latter forms (after the non-associative forms), the speaker had had enough time to settle on one specific strategy. Note also AK1’s systematic denasalization of η to g after n in forms of $múnún\eta$ ‘to give birth’ having undergone V2 deletion: $/múnún\eta-/ \rightarrow mún\eta\eta- \rightarrow múnúgú-$. I have not observed this denasalization

Table 5: CV.CVC verb roots with C-initial plural object suffixes (140318-23-AK1, 140318-23-AK1)

	1PL.EX.OBJ /-nùrú/ ~/-nũ/~/-rũ/	1PL.IN.OBJ /-nǎŋ/ ~/-rǎŋ/	2PL.OBJ /-nũŋ/ ~/-rũŋ/	3MF.PL.OBJ /-nìrí/ ~/-nĩ/~/-rĩ/
/jùgár/				
a. V2 del.	jùgr-nũ jùgr-rũ	—	—	jùgr-nĩ jùgr-rĩ
b. V2 & C _b del.	jùg-rũ	jùg-rǎŋ	jùg-rũŋ	jùg-rĩ jùg-nìrí
c. V2 & C _c del.	jùgr-ũ	jùgr-ǎŋ	jùg-rũŋ	jùgr-ĩ
d. V2 del. & V ep.	—	—	—	—
/págàr/				
a. V2 del.	pógr-rũ	—	—	—
b. V2 & C _b del.	póg-rũ	póg-rǎŋ	póg-rũŋ	póg-rĩ
c. V2 & C _c del.	pógr-ũ	pógr-ǎŋ	pógr-ũŋ	pógr-ĩ
d. V2 del. & V ep.	—	—	—	—
/múnúŋ/				
a. V2 del.	—	—	—	—
b. V2 & C _b del.	mún-nùrú	—	—	mún-nìrí mún-rĩ
c. V2 & C _c del.	múng-ũ	múng-ǎŋ	—	—
d. V2 del. & V ep.	múngù-nùrú	—	—	—

Table 6: CV.CVC verb roots with C-initial object+associative portmanteau suffixes (140318-23-AK1, 140318-23-AK1)

	1SG.OBJ+ASS /-ní/~/-rí/	2SG.OBJ+ASS /-nuá/~/-ruá/	3M.SG.OBJ+ASS /-ná/~/-rá/	3F.SG.OBJ+ASS /-nó/~/-ró/
/jùgàr/				
a. V2 del.	—	—	—	—
b. V2 & C _b del.	—	—	—	—
c. V2 & C _c del.	—	—	—	—
d. V2 del. & V ep.	jùgùr-ní	jùgùr-ná	jùgùr-ná	jùgùr-nú
/págàr/				
a. V2 del.	—	—	—	—
b. V2 & C _b del.	pág-rí	—	—	—
c. V2 & C _c del.	págr-í	—	—	—
d. V2 del. & V ep.	págór-ní	—	—	—
/múnún/				
a. V2 del.	—	—	—	—
b. V2 & C _b del.	—	—	—	—
c. V2 & C _c del.	múng-í	múng-á	múng-á	múng-ú
d. V2 del. & V ep.	múngú-ní	—	—	—

in anyone else's speech, even in effabl forms. For example, for /múnún-én/ 'give birth to me' AK1 says *múngîn*, while everyone else says *múnîñ*.

As shown in Table 7, all four possible strategies either violate otherwise inviolable phonotactic constraints (*CCC and the maximal disyllabicity of stems $*_{>\sigma\sigma}$), or delete segments that normally cannot be deleted, i.e., anything but V2 (represented as "No illicit deletion" in the table; violation of a constraint is indicated with a star, non-violation with a ✓).

Table 7: Violation of phonotactic constraints by CV.CVC-CV forms

		*CCC	* $_{>\sigma\sigma}$	No illicit deletion
a. V2 deletion:				
/jùgàr-ní/	→ jùgr-ní	*	✓	✓
b. V2 & C _b deletion:				
/jùgàr-rí/	→ jùg-rí	✓	✓	*
/múnún-rí/	→ mún-rí	✓	✓	*
c. V2 & C _c deletion:				
/jùgàr-rí/	→ jùg-rí	✓	✓	*
/múnún-nă/	→ múng-ă	✓	✓	*
d. V2 deletion & epenthesis:				
/jùgàr-ná/	→ jùgùr-ná	✓	*	✓
/múnún-ní/	→ múngú-ní	✓	*	✓

4.2 Frequency of ineffable forms

As could be expected, ineffability affects morpheme combinations that are rather rare. CVCVC and CVCCV verbs are indeed not frequent: they represent a total of 73 verbs (51 and 22 verbs respectively), i.e., 15% of the 486 verbs in my lexicon. Only 26 are transitive, i.e., compatible with object suffixes (although some of the intransitive ones are change-of-state verbs that can easily be transitivized through zero conversion). 14, i.e., 20% of these 73 verbs are recent loanwords, e.g., /págàr/ ‘to think (of)’ (from Chadian Arabic *fakkar*), /pilmé/ ‘to film’ (from French *filmer*).

In addition, the consonant-initial suffixes are limited to 1PL.INCL, 1PL.EXCL, 2PL, and 3MF.PL object suffixes, as well as all ventive and instrumental fused forms, which are very rarely used in my corpus. Opportunities for speakers to use transitive (or transitivized) CVCVC or CVCCV verbs with a consonant-initial suffix are thus relatively rare. Indeed, I have not found any of these verbs used with an object suffix in my corpus of spontaneous texts. However, speakers have no trouble producing such forms in elicitation when they are effable (i.e., with a V-initial suffix, e.g., /jùgàr-én/ → *jùgr-în* ‘shake-me’), and giving contexts in which their use would be natural.

5 An analysis of phonotactic and prosodic well-formedness in Laal

In this section, I give an account of the phonotactic rules and constraints discussed above, couched in classical Optimality Theory (Prince & Smolensky 1993, 2004). This choice of theoretical framework stems from the fact that I think it offers a good formalization of phonotactic constraints in general, and, more specifically, of the interesting aspects revealed by the Laal speakers’ behavior when faced with ineffability. The point of this paper is not to defend this approach against any other, but to use it as the heuristic tool that it is.

I first propose an account of regular V2 prevocalic gliding and deletion. Both apply to comply with the disyllabic maximum constraint, i.e., to avoid a violation of $*_{>\sigma\sigma}$, penalizing any output stem longer than two syllables. This markedness constraint is ranked higher than the faithfulness constraints MAX, penalizing any segment deletion (violated by V2 deletion) and IDENT[syll], penalizing any change to the feature [syllabic] (violated by V2 gliding). To account for the choice between prevocalic gliding and deletion, one more markedness constraint is needed: the undominated inventory-defining constraint $*_{\underline{V}}$, penalizing non-syllabic vowels (other than high front and high back vowels, which can be changed into the attested glides /j/ and /w/). These constraints are further defined and summarized in (24).

- (24)
- a. $*_{>\sigma\sigma}$: No output form should be longer than two syllables. One violation is incurred by any output form that has more than two syllables.
 - b. $*_{\underline{V}}$: No output form should contain non-syllabic vowels. One violation is incurred by every non-syllabic vowel in an output form (NB: glides *j* and *w* do not incur a violation)
 - c. MAX: No output form should be missing a segment present in the input. One violation is incurred by every segment in the input that is missing from the output.
 - d. IDENT[syll]: No output segment should have a different value for the feature [syllabic] than its input correspondent. One violation is incurred by every output seg-

ment whose [syllabic] feature specification is different from its input correspondent.

Since prevocalic gliding is preferred whenever possible, V2 deletion must be seen as a last resort repair. This is accounted for by ranking IDENT[syll] lower than MAX—the penalty for deletion of a full segment is higher than that for changing a single featural specification. The ranking is shown in (25), and illustrated with a V-final root combining with a V-initial suffix in the tableaux in (26) and (27), showing the mechanism behind V2 prevocalic gliding and V2 deletion respectively. Note that only repairs applied to V2 are shown here. V2 weakness is accounted for in tableaux (29)–(31) below.¹⁸

(25) $*_{>\sigma\sigma}, *\underline{V} \gg \text{MAX} \gg \text{IDENT}[\text{syll}]$

(26)

/pə̀dì-àn/	$*_{>\sigma\sigma}$	$*\underline{V}$	MAX	IDENT[syll]
a. pàdì.àn	*!			
→ b. pàdjàn				*
c. pàdàn			*!	

(27)

/gùmə-ér/	$*_{>\sigma\sigma}$	$*\underline{V}$	MAX	IDENT[syll]
a. gùmə.ír	*!			
b. gùməír		*!		*
→ c. gùmír			*	

As can be seen, V2 gliding of a high front vowel as in candidate (26b) does not violate $*\underline{V}$ or $*_{>\sigma\sigma}$, making it a more optimal repair than V2 deletion seen in suboptimal candidate (27c). Prevocalic gliding of a vowel other than /i/ or /u/, as in candidate (27b), violates $*\underline{V}$, which is ranked higher than MAX and thus makes candidate (27c), where V2 deletion has applied, more optimal.

We saw that when a C-final root combines with a V-initial suffix (or a V-final root with a C-initial suffix), V2 gliding is impossible because it would create a triconsonantal cluster, and V2 deletion applies systematically instead. We also saw the ban on clusters made up of more than two consonants is a direct consequence of syllable well-formedness constraints. These include $*\text{CPLXONSET}$ and $*\text{CPLXCODA}$, violated by complex onsets and complex codas respectively, $*\text{C}_{[+\text{syll}]}$, penalizing syllabic consonants, unattested in Laal, and PARSE , penalizing segments not parsed into a syllable, all undominated. I choose to merge these constraints into one general $*\sigma\text{-STRUCTURE}$ constraint ($*\sigma\text{-STR}$ for short), to save space in tableaux. This is illustrated in (28): prevocalic gliding in candidate (28b) violates $*\sigma\text{-STR}$, which is ranked higher than MAX, making candidate (28c) with V2 deletion the optimal candidate.¹⁹

¹⁸The faithful candidate in (26) and (27) lacks an onset consonant, and thus also violates undominated $*\sigma\text{-STRUCTURE}$, which accounts for syllable structure well-formedness, as defined below. Morphophonological changes involved in the output forms that are unrelated to well-formedness and ineffability are systematically applied and taken for granted, e.g., vowel harmony, regular tonal changes, etc.

¹⁹Candidate (28c) [múnwɛ́ɪn] can be understood to stand for several candidates violating various syllable-structure constraints subsumed under $*\sigma\text{-STR}$, including [mún.wɛ́ɪn] with a complex onset, [múnw.ɛ́ɪn] with a complex coda, [mún.{w}.ɛ́ɪn] with an unsyllabified consonant *w*, and [mún.wɛ́.ɪn] with a syllabic [ɪ] (which would additionally violate $*_{>\sigma\sigma}$).

(28)

/múnúŋ-én/	*> $\sigma\sigma$	* σ -STR	* $\check{V}$	MAX	IDENT[syll]
a. múnúŋín	*!				
b. múnwŋín		*!			*
→ c. múnŋín				*	

V2 weakness can be seen as emergent from the constraints on syllable structure subsumed under * σ -STR and the necessity to preserve inflectional and/or derivational information from the suffix. The latter can be accounted for by a positional faithfulness constraint *à la* Beckman (1998): FAITH-SFX, penalizing any change to suffixal material.²⁰ As seen in the tableau in (29), the ranking of FAITH-SFX above MAX explains why the deleted vowel in combinations of a V-final root and a V-initial suffix is the root-final vowel (as in candidate (29c)), and not the suffix vowel (as in candidate (29b)).²¹

(29)

/gùmà-ér/	*> $\sigma\sigma$	FAITH-SFX	MAX	IDENT[syll]
a. gùmà.ír	*!			
b. gùmár		*!	*	*
→ c. gùmír			*	

Constraints on syllable structure further explain why it is the root's second vowel rather than its first vowel that is deleted in order to reduce the number of syllables to two. This is seen in (30), where deletion of the vowel of the first syllable in (30b), while successfully avoiding a violation of *> $\sigma\sigma$, incurs a fatal violation of * σ -STR because of the complex onset it creates, thus making V2 deletion (candidate (30c)) a more optimal repair. Fixing the violation of * σ -STR by deleting one of the two consonants in order to simplify the complex onset incurs an additional violation of MAX, rendering this repair suboptimal, as illustrated by candidates (30d-e). Indeed, V2 deletion in candidate (30e) incurs only one such violation. This requires that violations of MAX be evaluated gradiently (as in the definition given in (24d)). Prevocalic gliding of V1 is of course not an option either, since it would create a triconsonantal cluster, fatally violating * σ -STR (candidate (30f)). Note that a positional faithfulness constraint protecting the initial syllable, e.g., MAX- σ 1, would also account for this, but would be redundant, since this is already accounted for by the constraints on syllable structure, independently required. I will come back to this in the next section.

²⁰This constraint subsumes several specific constraints, including IDENT-SFX penalizing featural changes (e.g., Prevocalic gliding) in suffixal material and MAX-SFX penalizing segmental deletion within a suffix. Note that only deletion and prevocalic gliding will be considered here; some changes to suffixal material are attested, in particular perseverative [+high] harmony, which I do not account for here, for simplicity. These would be easy to accommodate by fine-tuning the suffix-faithfulness constraints at work.

²¹The preference for prevocalic gliding of the root V2 rather than the suffix vowel cannot be illustrated because there are no V-initial suffixes starting with /i/ or /u/.

(30)

/jùgàr-én/	* _{>} σσ	*σ-STR	*V	MAX	IDENT[syll]
a. jùgàr-én	*!				
b. jgàr-én		*!		*	
→ c. jùgrén				*	
d. gàr-én				*!*	
e. jàr-én				*!*	
f. jwgàrén		*!			*

Gradient violation of MAX also accounts for the fact that only V2 is deleted to shorten a trisyllabic stems, and not a bigger portion or all of the second syllable. This can be seen in (31), where candidates (31c-d), in which both V2 and one of its adjacent consonants are deleted, violate MAX twice. Simple deletion of V2, on the other hand, violates MAX only once, making candidate (31b) optimal.

(31)

/jùgàr-én/	* _{>} σσ	MAX
a. jùgàr-én	*!	
→ b. jùgrén		*
c. jurén		*!*
d. jùgrén		*!*

The MAX ≫ IDENT[syll] ranking and the gradient evaluation of MAX violations models quite aptly the parsimonious nature of the repair strategies at work in avoiding stems longer than two syllables. The change should be the most minimal: change of [+syll] to [-syll] of one of the vowels whenever possible, i.e., if it is /i/ or /u/ and its prevocalic gliding does not violate syllable well-formedness; otherwise, deletion of a vowel. The fact that V2 is always the targeted vowel appears to be emergent from constraints on syllable structure: prevocalic gliding or deletion of V1 always creates an ill-formed syllable that requires further deletion, i.e., additional, unnecessary markedness violations.

6 Ineffability as a window into hidden grammar

The speakers' response to ineffability is interesting for at least two reasons. First, it seems to indicate that there is a threshold of acceptability or optimality, which cannot be represented in OT. It is as if a line were drawn at one of the ranking levels, and no candidate violating any constraint higher than this level can be deemed optimal, irrespective of lower violations. There is no such threshold in standard OT, where there is always an optimal candidate (or several tied optimal candidates in the case of free variation), even if it violates many high-ranked constraints.

A second point of interest is that, as already mentioned, ineffable forms in Laal are not fully ineffable: they are “effable under duress”, i.e., speakers accept to exceptionally “improvise” and venture outside of the “lawful” phonological grammar of their language in order to produce forms that are deemed suboptimal. This indicates that outputs violating constraints above the threshold, despite being deemed suboptimal, may still be considered when no other choice is available. However, only a subset of these suboptimal forms are considered, other possible repairs being strictly avoided (e.g., V1 is never deleted). There is thus some order—i.e.,

grammar—in the “illicit” operations that Laal speakers allow themselves to perform in order to solve ineffability. The threshold mentioned above can actually be seen as a double threshold, distinguishing between three types of constraints: the ones that are never violated by an optimal or tolerated suboptimal output form (the highest ranked), the ones that are never violated by an optimal output, but may be violated by a tolerated suboptimal output to solve ineffability (intermediate category between the higher and lower threshold), and the ones that are regularly violated by optimal outputs in effable cases, ranked below the lower threshold.²²

The speakers’ reponse to the issue of ineffability thus reveals aspects of the phonological grammar that are undetectable if one looks only at effable forms. An analysis based only on these effable forms, like the one developed in §5, misses this hidden grammar, and, I contend, mischaracterizes the phonology of Laal. In order to see this, let us start by applying the current analysis to the ineffable form /jùgár-nĩ~rĩ/ (where *nĩ~rĩ* indicates that both forms are attested). The result, which markedly differs from the speakers’ response to ineffability described above, is shown in tableau (32) (arrows in parentheses indicate candidates falsely predicted to be optimal by the constraint grammar).

(32)

	/jùgár-nĩ~rĩ/	*>σσ	*σ-STR	FAITH-SFX	MAX
faithful	a. jùgár-nĩ~rĩ	*!			
V2 del	b. jùgr-nĩ~rĩ		*!		*
V2 & C _b del	→ c. jùg-nĩ~rĩ				**
V2 & C _c del	d. jùgr-ĩ			*!	**
V1 del	f. jgàr-ni		*!		*
V1 & C _a del	(→) g. gàr-ni				**
V1 & C _b del	(→) h. jàr-ni				**

As can be seen, V2 deletion—the last-resort solution to a violation of *>σσ, as we saw—is not an optimal repair in this case, because it creates a triconsonantal cluster, violating undominated *σ-STR (candidate(32b)). Several further changes can be applied to reduce this triconsonantal cluster. With the constraint grammar defined above, there are three tied outputs: one in which the root-final consonant (i.e., the second consonant in the triconsonantal cluster created by V2 deletion, C_b) is deleted in addition to the second vowel (candidate (32c)), which is indeed one of the possible tolerated forms; and two candidates in which V1 is deleted instead of V2, together with one of the two adjacent consonants: the one before (the root-initial consonant) in candidate (32g), or the one after (the second consonant of the root) in candidate (32h). As already shown in (30), deletion of only V1 creates a complex onset, violating undominated *σ-STR, as in candidate (32f)).²³

The tableau in (32) is not representative of the speakers’ behavior described in §4 above. First, candidate (32d) with V2 and C_c deletion should be tied with candidate (32c), since it is one of the suboptimal candidates considered by speakers to solve ineffability. It is not the case here, since candidate (32d) with C_c deletion fatally violates FAITH-SFX, ranked above MAX. Secondly,

²²Outside of a constraint-based account, these two thresholds could be recast in terms of three types of processes: lawful processes (V2 deletion and prevocalic gliding), unlawful processes that are only exceptionally tolerated (C_b or C_c deletion, V epenthesis), and fully unlawful processes which are never resorted to (gliding or deletion of anything preceding V2).

²³In this section I only consider candidates where prevocalic gliding is not an optimal repair and ignore IDENT[syll] for simplicity.

V1 deletion (with or without deletion of one of the adjacent consonants) is never considered a possible solution to the ineffability issue, as we saw above (cf. (21)). Candidates (32g) and (32h) should thus be less optimal than (32c), contrary to what is shown in (32). Finally, one of the strategies applied by speakers in ineffable cases is not accounted for by the current constraint grammar, namely V2 deletion and V2 epenthesis (21d). The constraint grammar must thus be changed to reflect these hidden aspects of the phonology of Laal. This is done below.

The first change concerns the unavailability of V1 deletion (with or without further deletion of neighboring consonants) to reduce the number of syllables of the output form to a maximum of two. In the analysis presented in §5, it was simply viewed as emerging from general constraints on syllable structure. This analysis works only if V2 deletion solves all phonotactic violations (i.e., incurs no Markedness violations). When this is not the case, as with the ineffable forms in which V2 deletion creates a triconsonantal cluster, the analysis wrongly predicts, as shown in (32), that one of the “next best” suboptimal strategies should be deletion of V1 and one adjacent consonant, contrary to what the speakers actually do. The analysis must therefore explicitly rule out deletion of V1 and adjacent segments. This can be done, for instance, with a positional faithfulness constraint FAITH- σ 1 violated by any change to the initial syllable of the stem—a constraint which was deemed unnecessary in the analysis presented in §5, but which finds independent evidence in the stem-initial prominence manifested by the distributional asymmetries mentioned in §2.2 (cf. Tables 1 and 2). As seen in (33), FAITH- σ 1—which will be superseded by other constraints at a later stage of the analysis—takes care of eliminating candidates with V1 deletion (33f-h) from the tie. This leaves only the candidate with V2 and C_b deletion (33c) as optimal, which still does not fully reflect the speakers’ behavior described above.

(33)

	/jùgár-nĩ~rĩ/	*> $\sigma\sigma$	* σ -STR	FAITH-SFX	FAITH- σ 1	MAX
faithful	a. jùgár-nĩ~rĩ	*!				
V2 del	b. jùgr-nĩ~rĩ		*!			*
V2 & C _b del	→ c. jùg-nĩ~rĩ					**
V2 & C _c del	d. jùgr-ĩ			*!		**
V1 del	f. jgár-nĩ~rĩ		*!		*	*
V1 & C _a del	g. gár-nĩ~rĩ				*!	**
V1 & C _b del	h. jàr-nĩ~rĩ				*!	**

Another aspect that is not accounted for by the analysis above is the fact that C_a deletion, which would also simplify the triconsonantal cluster created by V2 deletion, is never resorted to by speakers as a means to solve ineffability. This is shown in the tableau in (34), where candidates (34c) and (34d) have exactly the same violation profile, and are consequently tied. Speakers actually prefer to delete the suffix-initial consonant (C_c), which our current constraint grammar clearly disprefers: candidate (34e) fatally violates FAITH-SFX, ranked above MAX, which is the only constraint violated by C_a deletion (34c).

(34)

	/jùgár-nĩ~rĩ/	*> $\sigma\sigma$	* σ -STR	FAITH-SFX	MAX
faithful	a. jùgár-nĩ~rĩ	*!			
V2 del	b. jùgr-nĩ~rĩ		*!		*
V2 & C _a del	(→) c. jùr-nĩ~rĩ				**
V2 & C _b del	(→) d. jùg-nĩ~rĩ				**
V2 & C _c del	e. jùgr-ĩ			*!	**

In fact, speakers never resort to deleting any of the segments preceding V2. This strongly suggests that V2 weakness is not just an emergent consequence of general constraints on syllable well-formedness, as in the analysis above. V2 seems to be the weakest segment in the stem, followed by post-V2 material whose deletion is exceptionally considered under ineffability, followed by pre-V2 material, which is never deleted. This is a crude generalization, which I tentatively formalize with the following ad-hoc constraints: $\text{MAX-preV2} \gg \text{MAX-postV2} \gg \text{MAX-V2}$. These are illustrated in the tableau in (35), where violations of constraints above the lower threshold of optimality by possible suboptimal solutions to ineffability are indicated with an exclamation mark in parentheses. Note that these new constraints supercede both $\text{FAITH-}\sigma 1$ (now replaced in its function by MAX-preV2) and FAITH-SFX (redundant with MAX-postV2).

The double threshold mentioned above is represented in the tableau with two thick vertical lines. MAX-preV2 and $*>\sigma\sigma$ are in the first tier, above the higher threshold: no candidate with a violation of these constraints can ever be optimal, even in ineffable cases. MAX-V2 (and $\text{IDENT}[\text{syll}]$, cf. (36)) are in the third, lowest tier: optimal candidates regularly violate these constraints. The most interesting tier is the second, intermediate one, between the lower and higher threshold, consisting here of MAX-postV2 , $*\sigma\text{-STR}$, and FAITH-SFX . Candidates violating one of these constraints can never be considered optimal. They may, however, be reluctantly tolerated as a “suboptimal” solution to the ineffability problem (when there is no candidate with a better violation profile).

What makes this particularly interesting is the marginal, unexpected, “incorrect but tolerated” nature of the tied candidates, which in regular OT should be considered as optimal in (35) as the optimal candidates in effable cases. Note that this analysis clearly departs from standard OT and ventures into slightly out-of-the-box, experimental territory. The main point here is not to propose a new framework, but to use this ad-hoc modification of OT as a basis to propose a tentative formalization of what I think the speakers’ behavior reveals about Laal grammar.

(35)

		$*>\sigma\sigma$	MAX-preV2	MAX-postV2	$*\sigma\text{-STR}$	MAX-V2
	/jùgár-nĩ~rĩ/					
faithful	a. jùgár-nĩ~rĩ	*!				
V2 del	→ b. jùgr-nĩ~rĩ				*(!)	*
V2 & C_b del	→ c. jùg-nĩ~rĩ			*(!)		*
V2 & C_c del	→ d. jùgr-ĩ			*(!)		*
V2 & C_a del	e. jùr-nĩ~rĩ		*!			*
V1 del	f. jgàr-nĩ~rĩ		*!		*	
V1 & C_a del	g. gàr-nĩ~rĩ		*!*			
V1 & C_b del	h. jàr-nĩ~rĩ		*!*			

As seen in the tableau above, candidates (35b) with V2 deletion, (35c) with V2 and C_b deletion, and (35d) with V2 C_c deletion are all tied, as expected, violating only intermediate constraints whose violation is tolerated to solve ineffability, and none of the first-tier, strictly inviolable constraints. Any deletion of pre-V2 material (including C_a and any segment belonging to the initial syllable) fatally violates first-tier constraints, as in candidates (35e-h). The general

tableau in (36) shows how this constraint grammar, more complex than the one presented in §5, accounts for the data seen so far (with the exception of suboptimal candidates with V2 epenthesis, addressed below).

(36)

	$*_{>\sigma\sigma}$	$*_V$	MAX_{-preV2}	$MAX_{-postV2}$	$*_{\sigma-STR}$	MAX_{-V2}	$IDENT_{[syll]}$
a. /pèdì-àn/							
i. pàdiàn	*!				*		
→ ii. pàdjàn							*
iii. pàdiàn		*!					*
iv. pàdàn						*!	
v. pàdìn				*!			
b. /gùmà-ér/							
i. gùmà.ér	*!				*		
ii. gùmà.ér		*!					*
iii. gùmà.ér		*!					*
iv. gùmár				*!			
→ v. gùm.ér						*	
c. /múnúŋ-én/							
i. múnúŋ.én	*!						
ii. múnwŋ.én					*!		*
→ iii. múnŋ.én						*	
d. /jùgàr-én/							
i. jùgàr.én	*!						
ii. jgàr.én			*!		*		
→ iii. jùgr.én						*	
iv. gàr.én			*!*				
v. jàr.én			*!*				
vi. jwgàr.én					*!		*
vii. jur.én			*!			*	
viii. jùg.én				*!		*	
e. /jùgàr-nĩ~rĩ/							
i. jùgàrnĩ~rĩ	*!						
(→) ii. jùgrnĩ~rĩ					*(!)	*	
(→) iii. jùgnĩ~rĩ				*(!)		*	
(→) iv. jùgrĩ				*(!)		*	
v. jùrnĩ~rĩ			*!			*	
vi. jgàrnĩ~rĩ			*!		*		
vii. gàrnĩ~rĩ			*!*				
viii. jàrnĩ~rĩ			*!*				

The last candidate considered by speakers that has yet to be accounted for is the one with V2 deletion and V2 epenthesis: /jùgár-nĩ/ → jùgùrnĩ (cf. (23)). This is impossible to account for within the same constraint grammar. Indeed, $* > \sigma\sigma$ has to be inviolable (i.e., ranked in the first tier), but it must also be exceptionally violated by the tied suboptimal candidate with V2 epenthesis (ranked in the second tier, between the two thresholds). This is a ranking contradiction, the only way out which would be to prevent epenthetic vowels from “counting” as syllable nuclei violating $* > \sigma\sigma$, either by restricting $* > \sigma\sigma$ to be violated only by trisyllabic forms in which all nuclei are underlying vowels, or by making epenthetic vowels in output forms somewhat invisible to certain constraints. This seems like an ad-hoc move that is unlikely to bring any insight as to what motivates the Laal speakers’ preference for V2 deletion and (re-)epenthesis over the non-application of V2 deletion as a way to avoid a triconsonantal cluster. At this point, one reaches the limit of the merits of the OT-inspired implementation chosen in this section.²⁴

What the preference for V2 epenthesis seems to be telling us is that speakers systematically “follow protocol” by applying the regular grammar, i.e., the regular solution to violations of $* > \sigma\sigma$, which is V2 deletion. Only then do they try to solve the problem posed by this application of regular phonology (the creation of a triconsonantal cluster) by trying illicit operations that are deemed acceptable in exceptional circumstances. In some cases, these illicit operations (C_b or C_c deletion) do indeed solve both the initial (trisyllabic form) and intermediate (triconsonantal cluster) phonotactic problems; in some other cases (V2 epenthesis), the illicit repair fixes only the intermediate issue, reintroducing the initial one to which it is blind. Interestingly, it seems that in order to count as an exceptionally acceptable solution to the ineffable problem, the final form must systematically bear the traces of an attempt to solve the initial problem, i.e., to reduce the number of syllables from three to two—even when it actually fails to do so. This is shown by the fact that the faithful candidate /jùgár-nĩ~rĩ/ → jùgàrnĩ~rĩ is never deemed acceptable, while jùgùrnĩ (with V2 epenthesis), which has exactly the same violation profile, is exceptionally acceptable as a response to ineffability.

7 Discussion and concluding remarks

As far as the conditioning on ineffability is concerned, the pattern described in this paper is typologically unsurprising: it is a *bona fide* case of phonologically conditioned ineffability driven by what Inkelas (2014: 290) calls ‘output sensitivity’, i.e., phonotactic and prosodic well-formedness constraints on output forms preventing specific forms from being produced.

Some of the properties of the Laal pattern are, however, unusual. The most obvious one is that, contrary to most documented cases, Laal speakers usually do end up producing a form when asked to derive prosodically problematic forms. Ineffability is thus not insurmountable. Interestingly, however, fixing ineffable forms involves employing “illicit means”, i.e., applying phonological operations that are otherwise not seen elsewhere in the grammar, which the speakers even seem to apply only reluctantly, as if “under duress”. Ineffability seems to take speakers by surprise, and the “illicit” repair they end up choosing to apply is partly unpredictable.

²⁴This could potentially be modeled in a derivational approach similar to Harmonic Serialism (McCarthy 2007, 2010, McCarthy & Pater 2016), in which input harmony is improved one change at a time: V2 deletion, which improves the form’s harmony with respect to $* > \sigma\sigma$, applies first, followed by V2 epenthesis, which applies to break the triconsonantal cluster created by V2 deletion and thus improve the same form’s harmony with respect to $*\sigma$ -STR.

Building on observations by Albright (2003, 2009), Inkelas (2014: 298–299) writes that

[T]he phonological problem that is apparently insurmountable in the ineffability examples is not typically a very serious one. In many cases the language either tolerates the same phonological structure in other contexts or exhibits, elsewhere, the very phonological alternations that could repair the phonologically illicit form.

(...)

It is not generally the case that phonological ineffability results from phonological problems so severe that the language cannot recover from them.

Laal is quite an interesting counter-example to this generalization. Indeed, instead of having truly ineffable forms (i.e., paradigm gaps) that involve phonological problems that the grammar inexplicably does otherwise have the means to “fix”, Laal has forms that the “regular” grammar cannot fix, but that the speakers do end up fixing by “going rogue”.

Furthermore, many cases of phonologically conditioned ineffability have been found to involve some amount of morphological conditioning as well, as pointed out by Albright (2003, 2009) (cf. Inkelas 2014: 296–299). This is not the case in Laal ineffability, which appears to be purely phonologically conditioned. The fact that trisyllabic forms involving a disyllabic object suffix /-nùrú/ or /-nirí/ are not repaired by V2 deletion and seem to be “allowed” to violate $*_{>\sigma\sigma}$ should be regarded as due to the exceptional status of these two suffixes (in this and other respects, e.g., optional application of vowel harmony) rather than a form of morphological conditioning on ineffability.

Finally, although speakers seem indecisive when choosing a suboptimal output to settle on in order to solve the ineffability issue, this indecisiveness is not due to an indeterminacy in the system, contrary to what Albright (2009) has found in other cases. It is not that there are too many possible solutions, but really that there is no good solution.

I have argued that the very interesting fact that speakers do end up choosing one of the “least bad solutions” to solve the ineffability issue reveals aspects of the phonological grammar that would otherwise go unnoticed. Indeed speakers do not seem to improvise freely in their “illicit” attempts at producing ineffable forms. They follow specific rules and principles. One important aspect that is revealed is the clear tripartite “strength” distribution of segments within the stem, with V2 being the weakest, post-V2 material the second weakest, and pre-V2 material the strongest. This is in keeping with both stem-initial prominence effects seen in the language (in particular the fact that consonant and vowel inventories are markedly richer in the stem-initial C and V slots), and the cross-linguistic tendency to preserve grammatical information by protecting suffixal material from deletion. But, crucially, it cannot be equated with these two principles. Indeed, deletion of the onset of the second syllable of the root (C2) to simplify triconsonantal clusters (e.g., in /jùgár-nĩ~rĩ/ → V2 deletion: *jùgr-nĩ~rĩ* → C2 deletion: *jùr-nĩ~rĩ*) is not allowed and never resorted to, despite the fact that this consonant is neither stem-initial, nor part of the suffix, and should therefore be the least protected, weakest consonant in the form. The only label that includes both stem-initial material and C2 while excluding V2 is “pre-V2 material”. The emergency operations that speakers resort to in order to solve ineffability reveal this category as one to be recognized as such in the phonological grammar of the language. This means that the default status of V2 deletion in effable cases is not due simply to independent constraints on syllable structure that make V2 deletion the only option that does

not create an ill-formed syllable, or to strong faithfulness to the stem initial CV sequence (or syllable) and suffixal material.

Finally, the formalism that I used to account for this hidden grammar, although useful for heuristic and exposition purposes, is to a large extent unsatisfactory. Some of the constraints used are indeed rather crude and ad-hoc, and the double threshold of optimality in OT tableaux is clearly unorthodox and deserves to be recast into a more solid and generalizable formalism—something I leave for future research(ers). I hope that this attempt at modeling what I think the Laal speakers' response to ineffability reveals about their language has at least shed some light on the importance of paying attention to such marginal aspects of both grammar and speaker behavior.

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Abbreviations

ASS	associative
EX	exclusive
IN	inclusive
F	feminine
M	masculine
N	neuter
OBJ	object
PL	plural
SG	singular
<i>sp.</i>	species
<i>spp.</i>	species (several)
VEN	ventive

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Résumé

French summary goes here

Zusammenfassung

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